확률·랜덤프로세스 기말대비 6/7 (D1) · 1차 — Lecture 05 연속확률변수 Part 1

① 오답노트.md 어제 오답부터 재풀이 ② 아래 신규 문제는 '상한'(시간 부족 시 신규만 이월) ③ 채점: origin 풀이 / /solve-problem

떠올릴 개념

- PDF: $f \geq 0$, $\int f = 1$, $P(a \leq X \leq b) = \int f$. → "otherwise = 0" 항상 명시!
- Uniform U(a,b): $E=(a+b)/2$, $Var=(b-a)^2/12$ · CDF $F=\int_{-\infty}^{x} f$, $f=F'$
- Exponential: $f=\lambda e^{-\lambda x}(x \geq 0)$, $E=1/\lambda$, $Var=1/\lambda^2$, 무기억성 · Geo→Exp 극한

문항 목차 (총 11문항)

Problem 1

Choose a real number uniformly at random in the interval $[2, 6]$ and call it X.

a. Find the CDF of X, $F_X(x)$.
b. Find EX.

— 풀이 —

Problem 2

Let X be a continuous random variable with the following PDF

$$f_X(x) = \begin{cases} ce^{-4x} & x \geq 0 \\ 0 & otherwise \end{cases}$$

where c is a positive constant.

a. Find c.
b. Find the CDF of X, $F_X(x)$.
c. Find $P(2 < X < 5)$.
d. Find EX.

— 풀이 —

Problem 3

Let X be a continuous random variable with PDF

$$f_X(x) = \begin{cases} x^2 + \frac{2}{3} & 0 \le x \le 1 \\ 0 & otherwise \end{cases}$$

a. Find $E(X^n)$, for $n = 1, 2, 3, \cdots$.
b. Find the variance of X.

— 풀이 —

Problem 7

Let $X \sim Exponential(\lambda)$. Show that

a. $EX^n = \frac{n}{\lambda} EX^{n-1}$, for $n = 1, 2, 3, \cdots$;
b. $EX^n = \frac{n!}{\lambda^n}$, for $n = 1, 2, 3, \cdots$.

— 풀이 —

Problem 11

Let $X \sim Exponential(2)$ and $Y = 2 + 3X$.

a. Find $P(X > 2)$.
b. Find EY and $Var(Y)$.
c. Find $P(X > 2|Y < 11)$.

— 풀이 —

4.2.1 ● The cumulative distribution function of random variable X is

$$F_X(x) = \begin{cases} 0 & x < -1, \\ (x+1)/2 & -1 \le x < 1, \\ 1 & x \ge 1. \end{cases}$$

(a) What is $P[X > 1/2]$?

(b) What is $P[-1/2 < X \le 3/4]$?

(c) What is $P[|X| \le 1/2]$?

(d) What is the value of a such that $P[X \le a] = 0.8$?

— 풀이 —

4.3.2 The cumulative distribution function of random variable X is

$$F_X(x) = \begin{cases} 0 & x < -1, \\ (x+1)/2 & -1 \le x < 1, \\ 1 & x \ge 1. \end{cases}$$

Find the PDF $f_X(x)$ of X.

— 풀이 —

4.4.1 ● Random variable X has PDF

$$f_X(x) = \begin{cases} 1/4 & -1 \le x \le 3, \\ 0 & \text{otherwise.} \end{cases}$$

Define the random variable Y by $Y = h(X) = X^2$.

(a) Find $\mathrm{E}[X]$ and $\mathrm{Var}[X]$.

(b) Find $h(\mathrm{E}[X])$ and $\mathrm{E}[h(X)]$.

(c) Find $\mathrm{E}[Y]$ and $\mathrm{Var}[Y]$.

— 풀이 —

4.4.6 The cumulative distribution function of random variable V is

$$F_V(v) = \begin{cases} 0 & v < -5, \\ (v+5)^2/144 & -5 \le v < 7, \\ 1 & v \ge 7. \end{cases}$$

(a) What are $\mathrm{E}[V]$ and $\mathrm{Var}[V]$?

(b) What is $\mathrm{E}[V^3]$?

— 풀이 —

1.3.4 Indicate whether each statement is true or false.

(a) If $P[A] = 2\,P[A^c]$, then $P[A] = 1/2$.

(b) For all A and B, $P[AB] \leq P[A]\,P[B]$.

(c) If $P[A] < P[B]$, then $P[AB] < P[B]$.

(d) If $P[A \cap B] = P[A]$, then $P[A] \geq P[B]$.

— 풀이 —

1.5.1 ● Given the model of handoffs and call lengths in Problem 1.4.1,

(a) What is the probability $P[H_0]$ that a phone makes no handoffs?

(b) What is the probability a call is brief?

(c) What is the probability a call is long or there are at least two handoffs?

— 풀이 —